MAH QING TONG

AI / LLM ENGINEER

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PROFESSIONAL SUMMARY

AI/LLM Engineer at Zetrix AI and Computer Science (Data Science) graduate from **Monash University Malaysia**. Experienced in architecting production Agentic RAG workflows, multi-hop retrieval systems, and autonomous agent graphs. Hands-on expertise with LangGraph, LangChain, Model Context Protocol (MCP), Qdrant, and FastAPI, specializing in hybrid dense/sparse search, cross-encoder reranking, deterministic tool execution, and Human-in-the-Loop orchestration.

WORK EXPERIENCE

Zetrix AI | LLM Engineer — JUN 2024 – PRESENT

- **Enterprise AI Workflow Automation:** Developed and deployed Dify-based LLM workflows for enterprise SOP and knowledge-base retrieval, supporting live operations with grounded, citation-backed responses.
- **Section-Aware RAG Ingestion:** Built automated document processing pipelines converting complex unstructured PDFs into hierarchical Markdown, improving document consistency and retrieval recall by 28%.
- **Agentic RAG & Deterministic Execution:** Architected the i-FIKR Islamic knowledge agent over 870+ records; integrated vector search with custom Python/Pandas sandbox tools to eliminate tally hallucinations and ensure exact filtering.
- **Multi-Hop Retrieval & Sub-Chunk Citations:** Engineered two-stage retrieval combining catalog-level metadata grounding with surgical chunk retrieval, enabling precise cross-document comparisons.
- **Conversational Booking Agent:** Implemented interactive agent handling real-time availability checks, appointment scheduling, and automated Human-in-the-Loop (HITL) handoff.

NIQ (NielsenIQ) | Data Input Validation Intern — NOV 2024 – FEB 2025

- **Dataset Validation at Scale:** Processed and validated large-scale retail datasets across thousands of records per batch, maintaining strict data integrity for downstream analytics pipelines.
- **Rule-Based Anomaly Detection:** Applied automated validation techniques to flag catalog discrepancies, boosting data reliability across Malaysia and Vietnam regional teams.

TECHNICAL PROJECTS

Personal DeepAgent — Autonomous Scholarly Research System — LangGraph • Hybrid RAG • FlashRank • DuckDB • FastAPI

- Architected an autonomous research agent featuring two-stage hybrid retrieval (dense Chroma + sparse BM25 RRF) and **FlashRank CPU cross-encoder reranking**, eliminating candidate crowding across multi-paper comparisons.
- Engineered autonomous task decomposition (`todo_write`), virtual workspace filesystem scratchpads (`write_file`), deterministic DuckDB SQL catalog verification, and an automated Needle-in-a-Haystack recall evaluator.

Interactive Portfolio & Dify Cloud RAG Agent — FastAPI • SSE Streaming • WebGL • Dify API

- Built full-stack portfolio with an embedded CLI chat agent streaming real-time token responses over Dify Cloud RAG.
- Implemented Harness Engineering conventions with state ledger tracking, Architectural Decision Records, and a 3-tier resilient RAG fallback architecture.

AI Job Search & Resume Agent — LangChain • Model Context Protocol (MCP) • Qdrant • Hybrid Search

- Engineered a multi-agent LLM pipeline automating job retrieval, ranking, and customized email outreach via JSearch API.
- Implemented dense + sparse BM25 hybrid search with cross-encoder reranking in Qdrant for semantic resume-job matching.

TECHNICAL SKILLS

Agentic AI & Orchestration:	Agentic RAG, LangGraph, LangChain, Model Context Protocol (MCP), Tool Calling, Human-in-the-Loop (HITL), Dify, Prompt & Context Engineering
RAG & Information Retrieval:	Hybrid Search (Dense + Sparse BM25), Qdrant, ChromaDB, FlashRank Cross-Encoder Reranking, Multi-Hop Retrieval, Document Preprocessing, Needle Evaluation
Programming & Backend:	Python, FastAPI, TypeScript, JavaScript, SQL, REST API Architecture, Next.js, Server-Sent Events (SSE)

Data & Developer Tools: DuckDB, PostgreSQL, MongoDB, Git, Docker, Linux, Pytest, CI/CD (GitHub Actions)

EDUCATION

Bachelor of Computer Science in Data Science
Second Upper Honours
Monash University Malaysia • 2023 – 2025
Coursework: Deep Learning, Algorithms, Big Data, Visualization

AWARDS & LEADERSHIP

- **Awards:** Monash Coding League 2024, ExxonMobil DataWorks Challenge 2024
- **Leadership:** President of Diabolo Society (Foon Yew) • Monash Rock Climbing Club